CENG543

PROMETHEUS 2 🔥

An Open Source Language Model Specialized in Evaluating Other Language Models

Duru Çiçek

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What Problem Are We Solving?

We want to evaluate the quality of outputs produced by language models (LMs)
We can evaluate those outputs with
humans, proprietary models, and open-source models,
but these bring in their own issues.

Why not proprietary **models?**

Proprietary models *perform very closely to human evaluation*, are more **cost-effective,** and **time-efficient**. But:

01 Proprietary LMs are expensive

02 Lack of transpereny, since their training data is secret → unreproducible

03 They are not customizable

04 They can disappear or change behavior overnight

Why not use the current open-source **evaluator models?**

Open-source evaluator models bring in more flexibility, but current models lack certain things

They disagree with humans → They give scores that correlate poorly with human judgments (see Figure 1, page 1).

They can only do one format → They usually support either:
direct assessment (score from 1–5),
OR
pairwise ranking (choose better of A/B)
but not both.

They cannot use custom evaluation criteria → They mostly evaluate **helpfulness** and **harmlessness**
But real-world evaluations use **many criteria**:
factuality, relevance, clarity, tone, domain-specific correctness…

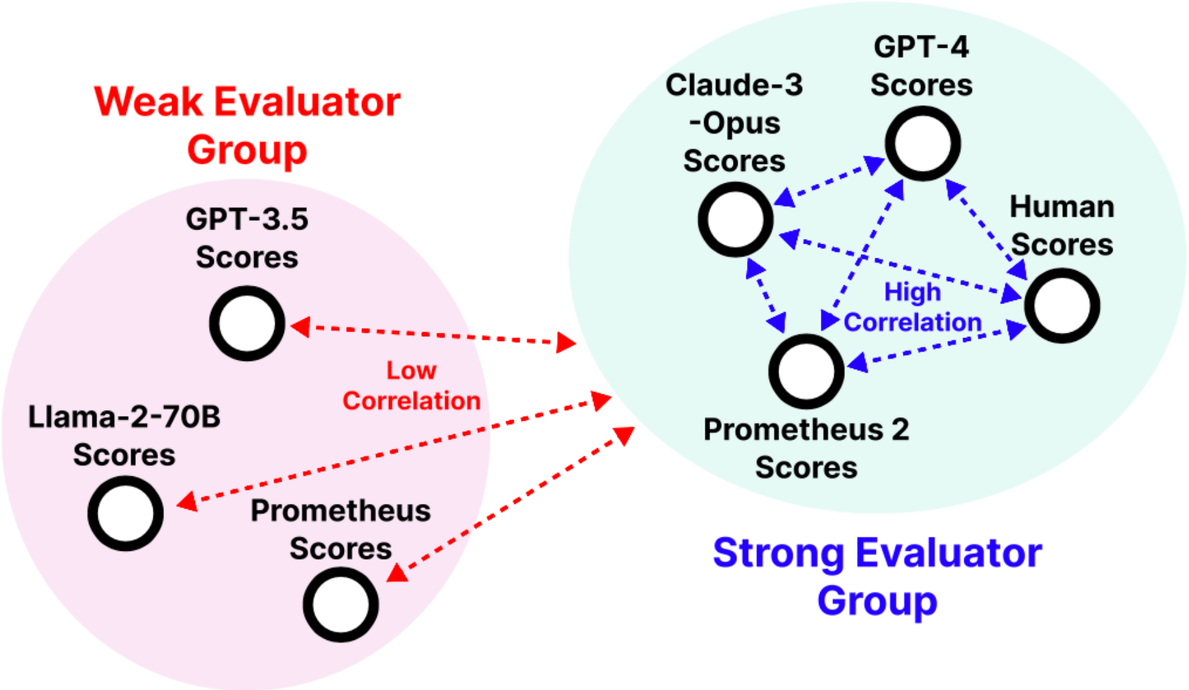

Figure 1: Weak evaluators (*e.g.*, Llama-2-Chat-70B, Prometheus, and GPT-3.5-Turbo) achieve low scoring correlation with strong evaluators (*e.g.*, Humans, GPT-4, and Claude-3-Opus). On the other hand, scores provided by strong evaluators highly correlate with each other.

Goal of the paper:

Build an open-source evaluator model as strong and flexible as GPT-4, able to perform BOTH direct scoring and pairwise ranking under ANY custom evaluation criteria.

How?

The key contribution is a unified training recipe:

1. Train **two evaluator models** separately
2. **Merge** them via task-level **weight merging**

This merged model can:

handle both pairwise ranking & direct assesment and match proprietary model performance more closely

Methodology

01

Direct Assesment

02

Pairwise Ranking

03

Preference Collection

04

Training Methods & Baselines

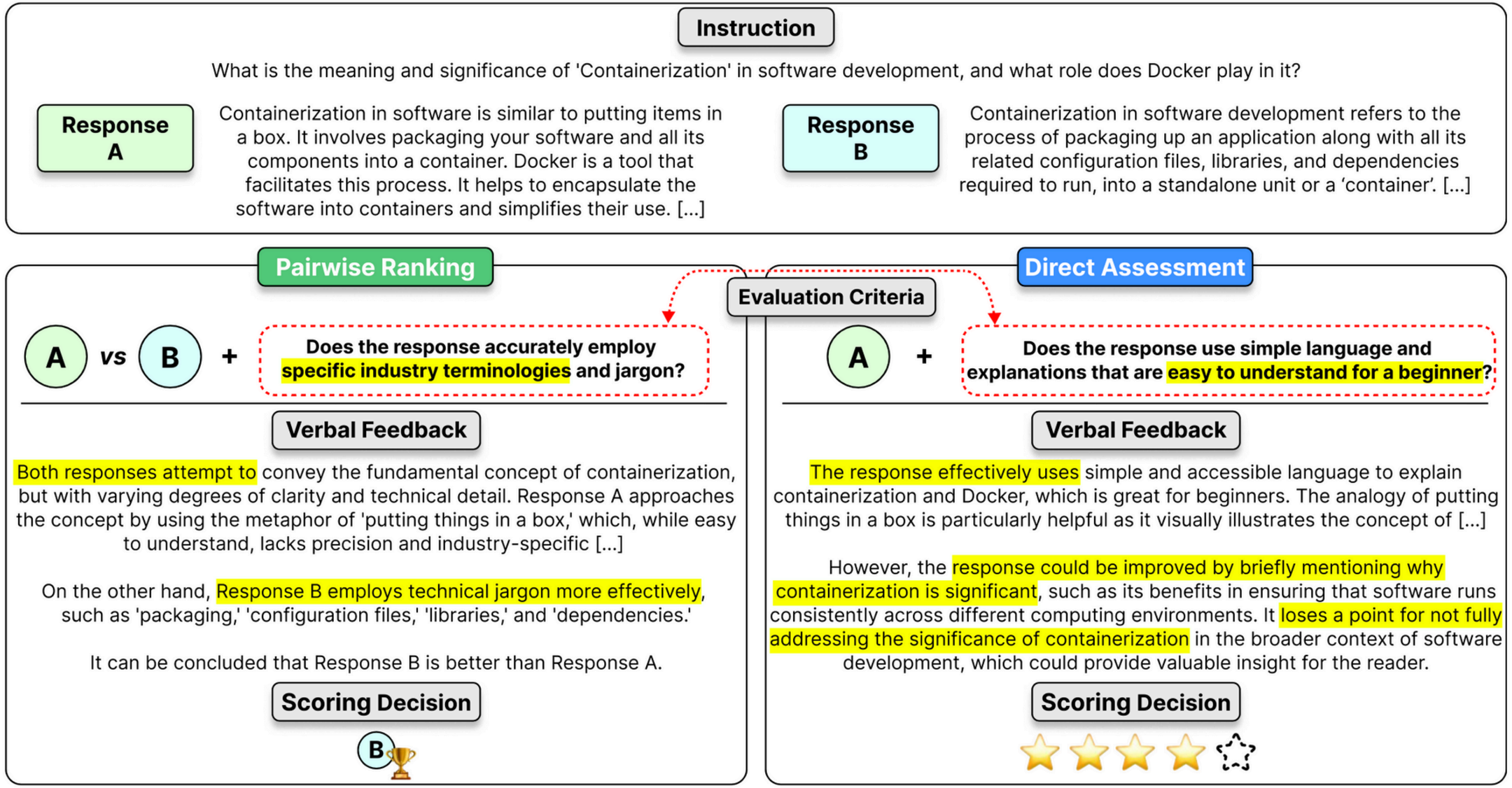

Figure 2: Comparison of direct assessment and pairwise ranking. Both responses could be considered decent under the umbrella of 'helpfulness'. However, the scoring decision might change based on a specific evaluation criterion.

Direct Assessment & Pairwise Ranking

$$f_{\text{direct}} : (i, r, a, e) \mapsto (v_r, s) \quad \text{where } s \in \{1, 2, 3, 4, 5\} \tag{1}$$

instruction *i*, response *r* →
scalar value score *s*,

*reference answer *a*
*verbal feedback *vr*
*score rubric with a criterion description and score-level descriptors *e*

$$f_{pair} : (i, r_m, r_n, a, e) \mapsto (v_{r_m, r_n}, s) \quad \text{where } s \in \{m, n\} \tag{2}$$

instruction *i*, responses *(rm, rn)* →
chosen pair s ∈ {m,n},

*reference answer *a*
*verbal feedback v_{r_m, r_n}
*score rubric with only a criterion description *e*

Dataset

They develop the **PREFERENCE COLLECTION**, a new fine-grained *pairwise ranking* feedback dataset that builds on the **FEEDBACK COLLECTION** (Kim et al., 2023), which is a *direct assessment feedback* dataset.

This is because popular pairwise ranking datasets do not include an evaluation criterion and verbal feedback.

Data	PREFERENCE COLLECTION	FEEDBACK COLLECTION
Evaluation Scheme	Pairwise Ranking	Direct Assessment
# Evaluation Criteria	1,000	1,000
# Instructions	20,000	20,000
# Reference Answer	20,000	20,000
# Instances	200,000	100,000
#Verbal Feedback	200,000	100,000

Table 1: Statistics of our training datasets, the FEEDBACK COLLECTION and the PREFERENCE COLLECTION. Note that the 1K evaluation criteria, 20K instructions, and 20K reference answers are *shared* among the two datasets. Both datasets have an equal number of scoring decisions ("A" or "B"; 100K each & 1-5; 20K each) to prevent unintended biases after training.

Training Methods & Baselines

1. Prompting (Zero-shot)

- No training, evaluate using only prompts
- Weak formatting & low correlation with humans

2. Single-Format Training

- Train **only on one format** (direct assessment or pairwise ranking)
- Strong within its own task, weak cross-task
- Examples: Prometheus-7B/13B, UltraRM, PairRM

Training Methods & Baselines

3. Joint Training

- Train one model on both formats simultaneously
- Causes negative task transfer → worse performance than single-format
- Struggles to balance absolute vs. relative evaluation

4. Weight Merging (PROMETHEUS 2)

- Train two specialists separately → merge weights
- Avoids conflicts seen in joint training
- Best performance across both scoring + ranking
- DARE-Linear merging works best

Baselines Compared:

- Llama-2-Chat models
- Mistral-7B, Mixtral-8×7B
- GPT-3.5, GPT-4, Claude-3 (proprietary references)

Evaluation Method	Benchmark	Metrics	Judgment Source	Reference Answer	# Score Rubrics	# Instructions	# Judgments
Direct Assessment	**Vicuna Bench**	Correlation	Proprietary LMs	Y	80	80	320
	MT Bench	Correlation	Proprietary LMs	Y	80	80	320
	FLASK	Correlation	Proprietary LMs & Humans	Y	12	200	2,000
	Feedback Bench	Correlation	Proprietary LMs	Y	200	200	1,000
Pairwise Ranking	**HHH Align.**	Accuracy	Humans	N	4	221	221
	MT Bench Human Judg.	Accuracy	Humans	N	1	80	3,360
	Auto-J Eval	Accuracy	Humans	N	1	58	1,392
	Preference Bench	Accuracy	Proprietary LMs	Y	200	200	2,000

Table 2: Statistics of our evaluation benchmarks to assess the evaluation capabilities of evaluator LMs.

Evaluator LM	VICUNA BENCH		MT BENCH		FLASK			Feedback Bench
	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	Humans	GPT-4-0613
LLAMA2-CHAT 7B	0.205	0.243	0.036	0.055	0.317	0.256	0.299	0.523
LLAMA2-CHAT 13B	0.185	0.141	-0.042	-0.002	0.239	0.247	0.263	0.545
LLAMA2-CHAT 70B	0.350	0.463	0.178	0.228	0.388	0.402	0.317	0.592
MISTRAL-INSTRUCT-7B	0.486	0.561	0.284	0.396	0.448	0.437	0.377	0.586
MIXTRAL-INSTRUCT-8X7B	0.566	0.579	0.551	0.539	0.483	0.495	0.420	0.673
PROMETHEUS-7B	0.484	0.528	0.378	0.382	0.352	0.331	0.348	0.847
PROMETHEUS-13B	0.492	0.534	0.404	0.477	0.462	0.470	0.449	0.860
AUTO-J (13B)	0.351	0.262	0.432	0.375	0.430	0.370	0.473	0.637
PROMETHEUS-2-7B	0.666	**0.654**	0.548	0.517	0.617	0.561	0.545	0.882
PROMETHEUS-2-8X7B	**0.685**	0.635	**0.665**	**0.614**	**0.659**	**0.626**	**0.555**	**0.898**
GPT-3.5-TURBO-0613	0.335	0.349	0.183	0.194	0.437	0.396	0.450	0.594
GPT-4-1106	/	0.694	/	0.717	/	0.736	0.679	0.753
CLAUDE-3-OPUS	0.694	/	0.717	/	0.736	/	0.573	0.788

Table 3: **Direct Assessment Results** Pearson correlations between reference evaluators (listed on top) and evaluator LMs. The best comparable statistics are **bolded** and second best underlined except proprietary LMs. Spearman and Kendall-Tau correlations are reported in Appendix C. Note that the Feedback Bench is an in-domain test set of the PROMETHEUS models.

Evaluator LM	HHH Alignment					MT Bench Human Judg.		Auto-J Eval		Preference Bench
	Help.	Harm.	Hon.	Other	Total Avg.	w/ TIE	w/o TIE	w/ TIE	w/o TIE	Instance-wise Criteria
Llama2-Chat 7B	55.93	62.07	49.18	62.79	57.01	46.68	50.39	45.76	45.73	58.60
Llama2-Chat 13B	71.19	77.59	60.66	62.79	68.33	51.22	49.61	47.84	43.28	63.00
Llama2-Chat 70B	62.71	81.03	65.57	65.12	68.78	**55.14**	60.88	53.38	50.64	64.70
Mistral-Instruct-7B	59.32	68.97	63.93	81.40	67.42	53.81	63.82	53.88	60.94	79.40
Mixtral-Instruct-8x7B	83.05	87.93	67.21	69.77	77.38	51.85	71.42	53.81	73.50	84.00
Pair RM (0.4B)	84.75	84.48	80.33	**90.70**	84.62	-	59.00	-	59.05	81.80
Ultra RM (13B)	**86.44**	79.31	**81.97**	88.37	83.71	-	56.00	-	59.85	86.97
Auto-J (13B)	77.97	79.31	70.49	74.42	75.57	42.56	69.12	43.46	76.64	81.35
Prometheus-2-7B	72.78	79.31	77.05	76.74	74.66	50.45	70.78	54.96	75.07	**93.25**
Prometheus-2-8x7B	84.75	**96.55**	**81.97**	76.74	**85.52**	55.07	**71.96**	**58.41**	**79.98**	90.65
GPT-3.5-Turbo-0613	77.97	81.03	77.05	67.44	76.47	54.65	69.41	45.98	72.13	75.05
GPT-4-1106-Preview	89.83	96.55	91.80	83.72	90.95	60.38	79.90	52.80	83.12	85.50
Claude-3-Opus	91.53	100.00	91.80	95.35	94.57	55.35	77.65	60.70	82.92	89.85

Table 4: **Pairwise Ranking Results** Accuracy on human preference datasets. The best comparable accuracies are **bolded** and second best underlined except proprietary LMs. Note that HHH Alignment is an in-domain test set for PairRM, Auto-J Eval is an in-domain test set for Auto-J, and the Preference Bench is an in-domain test set for Prometheus-2 models.

Analysis of the Weight Merging Technique

RQ1: Weight merging vs. joint training?

Joint training caused **negative task transfer** (where performance on a single task was worse than a model trained only for that task).
Weight merging, in contrast, shows **positive task transfer**.

RQ2: Is the effectiveness of weight merging due to model ensembling?

An experiment merging two models trained on the same format (direct a. + direct a.) showed *no improvement* and often *worsened* performance.

The benefit arises specifically from unifying models *trained on different evaluation formats*.

RQ3: To what extent does learning with direct assessment help pairwise ranking performance, and vice versa?

Analysis of the merging coefficient (α) reveals that knowledge transfer is **asymmetric**.
Training on *pairwise ranking data* provides a *more significant performance boost* to *direct assessment tasks* than the reverse.

Training Method	Direct Assessment Benchmarks				Pairwise Ranking Benchmarks			
	Vicuna Ben.	MT Ben.	FLASK	Average	HHH Align.	MT Ben. H.J.	Auto-J Eval	Average
Mistral-Instruct-7B								
Prompting	0.486	0.284	0.480	0.417	67.42	63.82	60.94	64.06
Direct Assessment Only	0.537	**0.561**	0.519	0.539	73.33	56.76	64.38	64.82
Pairwise Ranking Only	-	-	-	-	78.73	67.06	72.03	72.61
Joint Training	0.548	0.450	0.457	0.485	**80.09**	65.49	73.60	73.06
Weight Merging	**0.666**	0.548	**0.659**	**0.624**	74.66	**70.78**	**75.07**	**73.50**
Mixtral-Instruct-8x7B								
Prompting	0.566	0.551	0.507	0.541	77.38	71.42	73.55	74.56
Direct Assessment Only	0.625	0.664	0.587	0.625	74.21	53.14	65.85	64.40
Pairwise Ranking Only	-	-	-	-	84.16	66.27	75.66	75.36
Joint Training	0.628	0.560	0.596	0.595	82.35	68.73	74.78	75.29
Weight Merging	**0.685**	**0.665**	**0.659**	**0.670**	**85.52**	**71.96**	**79.98**	**79.15**

Table 5: **Single-Format Training vs Joint Training vs Weight Merging** Pearson correlations between evaluator LMs trained with different methods and GPT-4-1106. Evaluator LMs trained with weight merging outperform single-format-trained and jointly-trained evaluator LMs across multiple benchmarks.

Training Data Evaluation Format	Direct Assessment Benchmarks				Pairwise Ranking Benchmarks			
	Vicuna Ben.	MT Ben.	FLASK	Average	HHH Align.	MT Ben. H.J.	Auto-J Eval	Average
No Training (Prompting)	0.486	0.284	0.480	0.417	67.42	63.82	60.94	64.06
Direct Assessment Only	0.537	**0.561**	0.519	0.539	73.33	56.76	64.38	64.82
Pairwise Ranking Only	-	-	-	-	**78.73**	67.06	72.03	72.61
Direct Assessment & Direct Assessment	0.552	0.493	0.505	0.517	73.30	55.00	63.69	64.13
Pairwise Ranking & Pairwise Ranking	-	-	-	-	78.70	65.20	72.72	72.21
Direct Assessment & Pairwise Ranking	**0.666**	0.548	**0.659**	**0.624**	74.66	**70.78**	**75.07**	**73.50**

Table 6: **Unifying Formats vs Ensembling** Pearson correlations with GPT-4-1106 (Vicuna Bench, MT Bench, FLASK) and agreement with human evaluators (HHH Alignment, MT Bench Human Judgment, Auto-J Eval). Merging models trained with the same evaluation formats (ensembling) underperforms merging models trained with different formats (unifying formats).

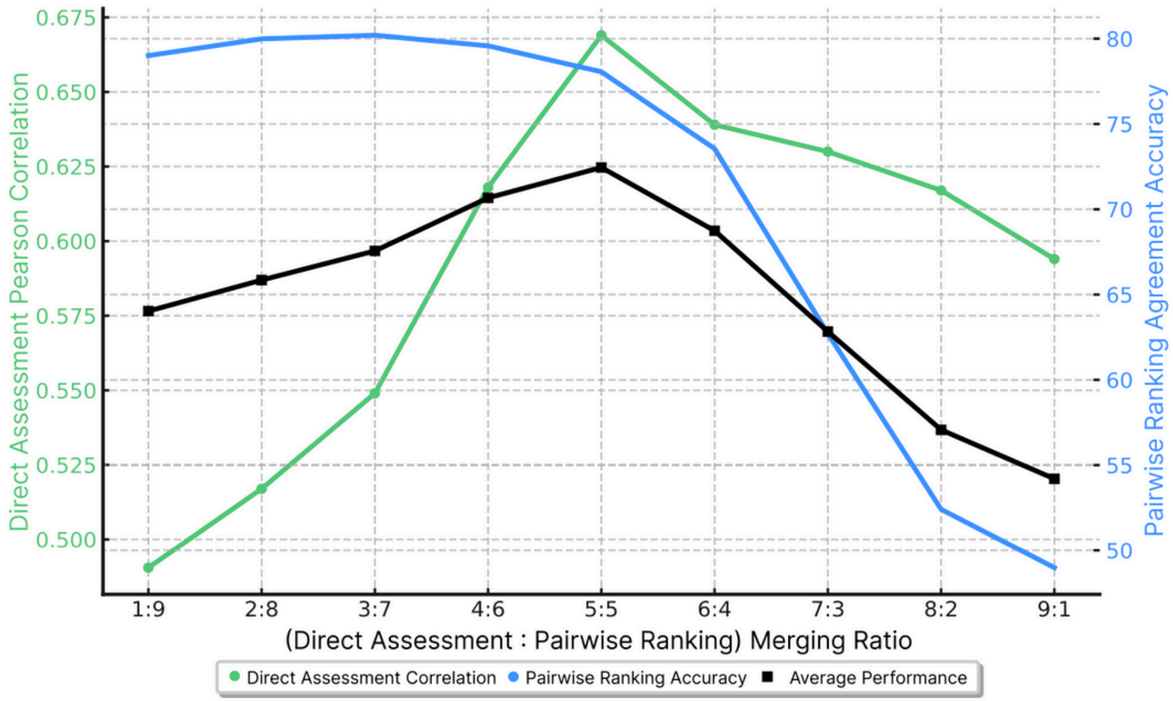

Figure 3: **When merging models, the influence of relative evaluation on absolute evaluation is greater than the influence of absolute evaluation on relative evaluation.** Performance of Direct Assessment (colored in green) and Pairwise Ranking (colored in blue) when altering the α value to merge evaluator LMs trained on different formats.

Limitations and Future Directions

- **Indirect Evaluation:** The study relies on correlation with human/proprietary judgments as a *proxy* for quality. Future work could explore more direct meta-evaluation pipelines.
- **Data Scarcity:** Scarcity of high-quality human judgment data for training and evaluating evaluator LMs.
- **Format Constraints:** It lacks the zero-shot flexibility of proprietary LMs, and other formats like 1-10 scales or ranking multiple responses at once.
- **Theoretical Understanding:** While empirically effective, the fundamental reasons why weight merging works so well in this context remain an open question for theoretical analysis.

THANK YOU